Stability Analysis and Linear Buckling of Structures: Beams, Trusses, and Thin Plates

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Abstract

This laboratory work evaluates linear elastic buckling (MSC Nastran SOL 105) alongside classical analytical solutions across three structural configurations. First, the critical loads of a slender tubular column are evaluated under clamped-free, pinned-pinned, and clamped-pinned boundary conditions to assess effective length factors. Second, a planar truss under asymmetric nodal loading is analyzed to identify the most heavily compressed member and its Euler buckling margin. Finally, a quarter-symmetric model of a simply supported thin plate under uniaxial compression is computed using work-equivalent nodal loads and symmetry constraints. Results are systematically compared against Euler-Bernoulli and Kirchhoff plate theories to assess model accuracy and geometric stiffness effects.

Keywords: *Finite Element Method, Linear Buckling, Structural Stability, MSC Nastran SOL 105, Euler Column, Differential Stiffness Matrix, Truss Instability, Thin Plate Buckling*

1. THEORETICAL FRAMEWORK: LINEAR BUCKLING FORMULATION

1.1. Physical Nature of Elastic Buckling

Structural stability problems differ fundamentally from standard linear static stress problems. While stress analysis focuses on equilibrium in the initial, undeformed configuration, stability analysis investigates the behavior of the equilibrium state when subjected to infinitesimal kinematic perturbations.

Under compressive stresses, a slender structure may experience a sudden transition from its primary equilibrium state (axial compression) to an adjacent, energetically favorable secondary equilibrium state involving large transverse deflections. This phenomenon, termed *bifurcation buckling*, often precipitates catastrophic structural collapse without prior warning and at stress levels well below the material yield strength ($\sigma_{\text{crit}} \ll \sigma_y$).

1.2. Euler Column Formulation and Boundary Conditions

The fundamental equation governing the elastic flexural buckling of an idealized, perfectly straight column was derived by Leonhard Euler. For a column of length L, modulus of elasticity E, and minimum second moment of area I, pinned at both ends, the critical axial load is expressed as:

$$P_{cr} = \frac{\pi^2 EI}{L^2} \tag{1}$$

To extend this formulation to arbitrary kinematic boundary conditions, the effective buckling length L_f is introduced via an effective length factor K:

$$L_f = K \cdot L \;\Rightarrow\; P_{cr} = \frac{\pi^2 EI}{(KL)^2} = \frac{\pi^2 EI}{L_f^2} \tag{2}$$

The parameter K accounts for rotational and translational edge restraints:

- **Clamped – Free (Cantilever):** $K = 2.0$, $L_f = 2.0L$
- **Pinned – Pinned (Ball Joint / Rotule):** $K = 1.0$, $L_f = 1.0L$
- **Clamped – Pinned:** $K \approx 0.7$ (exact: 0.699), $L_f = 0.7L$
- **Clamped – Clamped:** $K = 0.5$, $L_f = 0.5L$

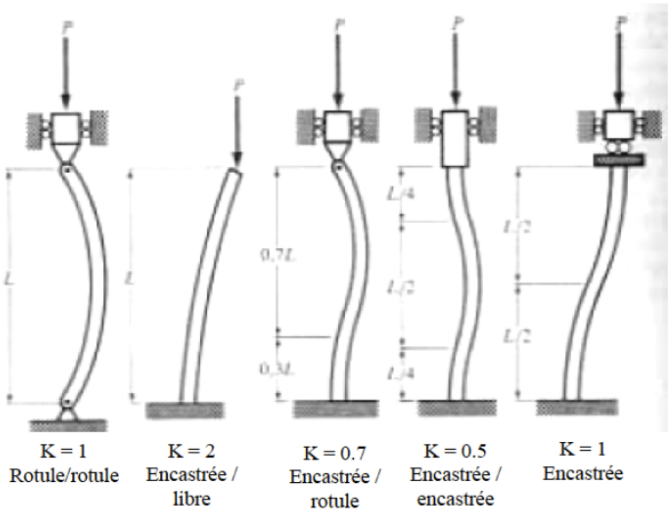


Figure 1. Buckling cases

1.3. Finite Element Eigenvalue Formulation (SOL 105)

In linear static analysis, the strain-displacement relations are assumed to be linear (infinitesimal strain assumption). However, in stability analysis, second-order terms of the Green-Lagrange strain tensor must be accounted for:

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} + \sum_k \frac{\partial u_k}{\partial x_i}\frac{\partial u_k}{\partial x_j}\right) \tag{3}$$

This second-order kinematic term gives rise to the *differential geometric stiffness matrix* $[K_D]$ (also denoted $[K_G]$), which depends directly on the internal stress distribution induced by an applied reference load P_a:

$$[K] = [K_E] + [K_D(P_a)] \tag{4}$$

where:

- $[K_E]$ is the conventional elastic stiffness matrix (governed solely by material elasticity and linear geometry).
- $[K_D]$ is the geometric stiffness matrix. In compression, $[K_D]$ is negative-definite, reducing the overall structural stiffness, whereas in tension, it stiffens the structure.

At the critical buckling load $P_{cr} = \lambda P_a$, the perturbed equilibrium requires that the total tangent stiffness vanishes, leading to the generalized eigenvalue problem:

$$([K_E] + \lambda[K_D])\{u\} = \{0\} \tag{5}$$

A non-trivial deformation mode $\{u\} \neq \{0\}$ exists if and only if:

$$\det([K_E] + \lambda[K_D]) = 0 \tag{6}$$

In MSC Nastran, this linearized buckling analysis is designated as **Solution 105 (SOL 105)**. The calculation proceeds in two automated stages:

1. **Subcase 1 (Linear Statics):** Applies the nominal reference load P_a to compute the internal element stress fields and construct $[K_D]$.
2. **Subcase 2 (Real Eigenvalue Extraction):** Employs the Lanczos algorithm to extract the eigenvalues λ_i and eigenvectors $\{u_i\}$. The critical buckling load is obtained from the lowest positive eigenvalue:
$$P_{cr} = \lambda_1 \cdot P_a \tag{7}$$

2. EXERCISE 1: BUCKLING ANALYSIS OF A COLUMN BEAM

2.1. Problem Statement and Section Properties

We consider a vertical tubular steel column with the following technical specifications:

- Total length: $L = 4.0$ m
- Outer diameter: $D = 20$ cm $= 0.20$ m
- Wall thickness: $e = 2$ cm $= 0.02$ m
- Inner diameter: $d = D - 2e = 0.16$ m
- Material: Structural steel with $E = 200$ GPa $= 200 \times 10^9$ Pa, $\nu = 0.3$, and yield limit $\sigma_{R0.2} = 600$ MPa $= 600 \times 10^6$ Pa.

The cross-sectional area S of the hollow circular tube is:

$$S = \frac{\pi}{4}(D^2 - d^2) = \frac{\pi}{4}(0.20^2 - 0.16^2) \approx 1.1310 \times 10^{-2} \text{ m}^2 \tag{8}$$

The area moment of inertia I with respect to any diametral axis is:

$$I = \frac{\pi}{64}(D^4 - d^4) = \frac{\pi}{64}(0.20^4 - 0.16^4) \approx 4.6370 \times 10^{-5} \text{ m}^4 \tag{9}$$

The flexural rigidity of the beam is:

$$EI = (200 \times 10^9) \times (4.6370 \times 10^{-5}) \approx 9.2740 \times 10^6 \text{ N} \cdot \text{m}^2 \tag{10}$$

2.2. Analytical Sizing: Yielding vs. Buckling

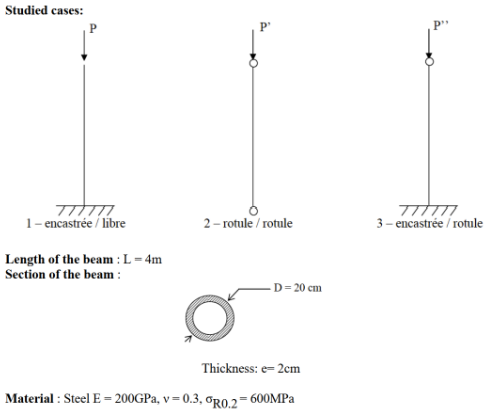


Figure 2. Study cases and data

Question 1.1: Maximum Force Without Plastic Deformation Assuming a perfectly rigid structure immune to geometric instability, structural capacity is solely limited by material yield under pure axial compression:

$$F_{\text{plastic}} = \sigma_{R0.2} \cdot S = 600 \times 10^6 \times 1.1310 \times 10^{-2} \approx \boxed{6.786 \text{ MN}} \tag{11}$$

The load $F_{\text{plastic}} = R_e \cdot S$ represents the theoretical full plastic yield capacity of the cross-section under pure axial compression. For a slender beam, this limit serves as an unrealistic theoretical upper bound: the structure will experience catastrophic failure via elastic instability (Euler buckling) at a critical load threshold well below this yield value ($P_{\text{cr}} \ll F_{\text{plastic}}$), long before the material reaches its yield strength. **This is mainly due to imperfection in the material.**

Question 1.2: Euler Critical Buckling Loads We evaluate the critical buckling load P_{cr} for the three prescribed boundary condition cases:

1. **Case 1: Clamped – Free ($K = 2, L_f = 8$ m):**
$$P_{cr,1} = \frac{\pi^2 EI}{(2L)^2} = \frac{\pi^2 \times 9.2740 \times 10^6}{64} \approx \boxed{1.430 \text{ MN}} \tag{12}$$
2. **Case 2: Pinned – Pinned ($K = 1, L_f = 4$ m):**
$$P_{cr,2} = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 \times 9.2740 \times 10^6}{16} \approx \boxed{5.721 \text{ MN}} \tag{13}$$
3. **Case 3: Clamped – Pinned ($K = 0.7, L_f = 2.8$ m):**
$$P_{cr,3} = \frac{\pi^2 EI}{(0.7L)^2} = \frac{\pi^2 \times 9.2740 \times 10^6}{7.84} \approx \boxed{11.675 \text{ MN}} \tag{14}$$

2.3. Finite Element Modeling in Patran / Nastran

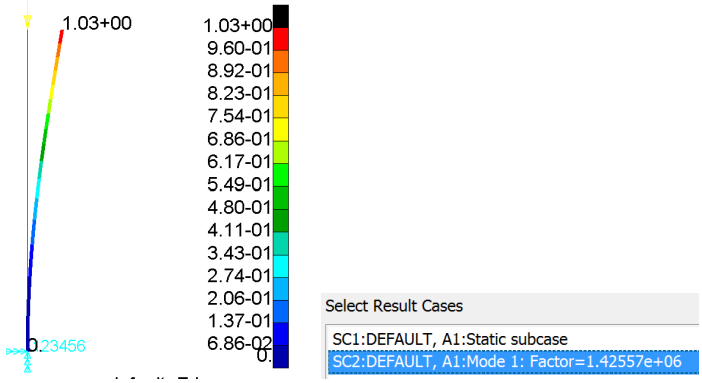
The beam is modeled vertically along the z-axis (or y-axis) using 1D `CBEAM` or `CBAR` elements:

1. **Geometry & Mesh:** A straight line curve of length $L = 4.0$ m is discretized into 20 beam elements to provide sufficient kinematic degrees of freedom to accurately capture the curvature of the first buckling mode.
2. **Material Properties:** Defined via `MAT1` card ($E = 200$ GPa, $\nu = 0.3$).

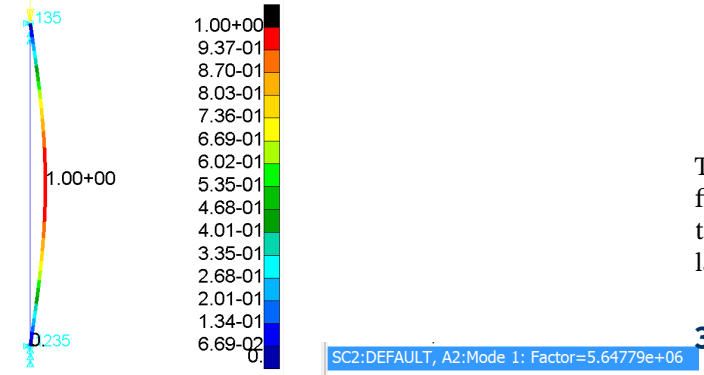
3. **Element Properties:** Assigned via PBEAM / PBAR using standard circular tube dimensions ($r_{\text{ext}} = 0.10$ m, $t = 0.02$ m).
4. **Boundary Conditions:** Applied at the base ($z = 0$) and tip ($z = 4$ m) according to the investigated case.
5. **Loading:** A unit compressive load $P_a = 1.0$ N is applied at the free or guided tip.
6. **Analysis Sequence:** Solution Type **Buckling (SOL 105)**, requesting OP2 and Print output formats, and Lanczos eigenvalue extraction for the primary mode.

Figure 3. Buckling mode shapes and eigenvalue factors.

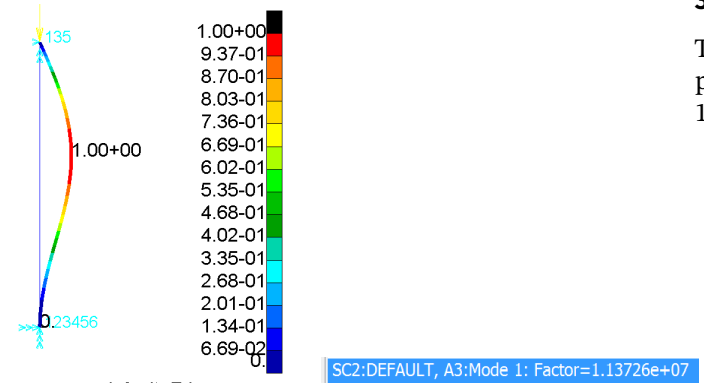
Case 1: Encastrée / Libre ($K = 2.0$)



Case 2: Rotule / Rotule ($K = 1.0$)



Case 3: Encastrée / Rotule ($K = 0.7$)



2.4. Question 1.3: Comparison and Error Analysis

The critical load obtained from finite element analysis (SOL 105) is directly given by:

$$P_{cr,\text{FE}} = \lambda_1 \cdot P_a \tag{15}$$

where λ_1 is the first extraction eigenvalue and $P_a = 1.0$ N is the nominal applied compressive load. The relative error with respect to the analytical Euler solution is determined as follows:

$$\text{Error (\%)} = \frac{P_{cr,\text{analytical}} - P_{cr,\text{FE}}}{P_{cr,\text{analytical}}} \times 100\% \tag{16}$$

Table 1. Comparison between analytical Euler buckling loads and MSC Nastran SOL 105 eigensolutions.

Condition	Factor K	$P_{cr,\text{ana}}$ [MN]	$P_{cr,\text{FE}}$ [MN]	Error (%)
Case 1: Encastrée / Libre	2.0	1.4302	1.4256	0.32
Case 2: Rotule / Rotule	1.0	5.7208	5.6478	1.28
Case 3: Encastrée / Rotule	0.7	11.6751	11.3726	2.59

The numerical eigensolutions demonstrate excellent correlation with theoretical Euler predictions across all three boundary conditions:

- **Case 1 (Encastrée / Libre):** Yields an eigenvalue of $\lambda_1 = 1.42557 \times 10^6$, corresponding to $P_{cr,\text{FE}} = 1.4256$ MN and a minimal discrepancy of 0.32%.
- **Case 2 (Rotule / Rotule):** Produces $\lambda_1 = 5.64779 \times 10^6$ ($P_{cr,\text{FE}} = 5.6478$ MN) with an error of 1.28%. Locking the drilling rotation degree of freedom (R_y) at the supports successfully eliminated the rigid-body pivot singularity without introducing artificial bending stiffness. The single half-wave mode shape reaches its normalized apex precisely at mid-span ($y = L/2$).
- **Case 3 (Encastrée / Rotule):** Obtains an eigenvalue of $\lambda_1 = 1.13726 \times 10^7$ ($P_{cr,\text{FE}} = 11.3726$ MN), showing a deviation of 2.59% relative to the standard engineering approximation ($K = 0.7$). The inflection point and deflection peak are correctly shifted toward the pinned end, reflecting the rotational clamping constraint ($R_x = R_z = 0$) at the base.

The slight, consistent reduction in stiffness observed across all finite element models compared to classical Euler-Bernoulli theory originates from the transverse shear flexibility formulated within MSC Nastran's beam elements

3. EXERCISE 2: BUCKLING ASSESSMENT OF A TRUSS STRUCTURE

3.1. Structural Configuration and Design Specifications

The studied structure is a symmetrical planar roof truss comprising 25 structural members, spanning a total distance of 18.0 m subdivided into 6 bays of 3.0 m each.

- End uprights height: $h_1 = 2.000$ m
- Central ridge height: $h_{\text{peak}} = 5.000$ m
- Boundary restraints: Pinned support (rotule) at left node 1, roller support (appui simple) at right node 7.
- Loading condition (applied downward at upper chord joints):
  - Outer joints: $F = 10$ kN
  - Intermediate joints: $F = 15$ kN
  - Apex ridge joint: $F = 20$ kN
- Total applied downward load:
$$F_{\text{total}} = 2 \times 10 + 4 \times 15 + 20 = 100 \text{ kN} \tag{17}$$
By global vertical equilibrium and symmetry, the reaction force at each support is $R_A = R_B = 50$ kN.
- Bar cross-section: Solid circular bar of diameter $D = 8$ cm $= 0.08$ m.

- Material properties: Steel with $E = 210\,000$ MPa $= 210$ GPa, $\nu = 0.3$.

3.2. Section and Geometric Properties

For all members of diameter $D = 0.08$ m:

- Cross-sectional area:
$$S = \frac{\pi D^2}{4} = \frac{\pi \times (0.08)^2}{4} \approx \boxed{5.0265 \times 10^{-3}\ \text{m}^2} \tag{18}$$
- Moment of inertia:
$$I = \frac{\pi D^4}{64} = \frac{\pi \times (0.08)^4}{64} \approx \boxed{2.0106 \times 10^{-6}\ \text{m}^4} \tag{19}$$
- Radius of gyration:
$$i = \sqrt{\frac{I}{S}} = \frac{D}{4} = \frac{0.08}{4} = 0.02\ \text{m} \tag{20}$$

3.3. Question 2.1: Finite Element Static Calculation

The truss is modeled in MSC Patran using 1D truss elements (CROD), or pin-ended beam elements (CBAR with end releases), so that only axial forces N are transmitted through frictionless pinned joints.

A linear static analysis (SOL 101) is performed to obtain the axial force in every bar and to identify the most compressed member ($N_{\max}$ in compression).

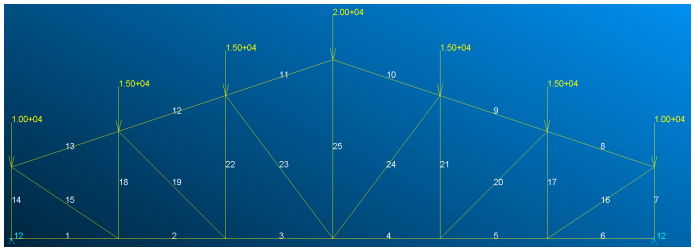


Figure 4. Finite element model of the planar truss with joint loads (in N) and support constraints.

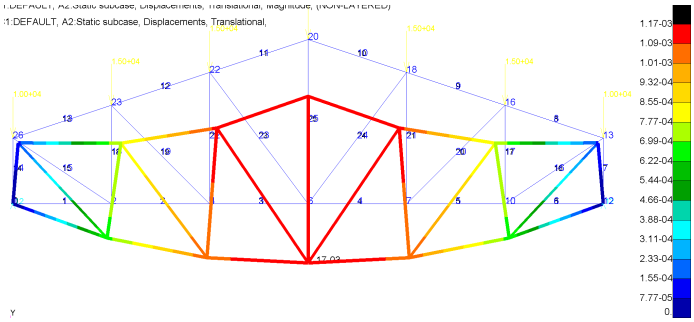


Figure 5. Translational displacement magnitude (amplified deformed shape) obtained from MSC Nastran SOL 101.

3.4. Identification of the Most Compressed Bar and Physical Interpretation

3.4.1. Numerical Results from MSC Nastran SOL 101

The linear static analysis of the constrained truss confirms that the model is in static equilibrium, without any kinematic mechanism:

- **Maximum deflection:** the peak displacement occurs at the mid-span bottom chord node (Node 6):
$$\delta_{\max} = 1.17 \times 10^{-3}\ \text{m} \quad (1.17\ \text{mm}), \tag{21}$$
which is consistent with the stiffness of a steel truss under the total load of 100 kN.
- **Most compressed bars:** the governing members are the inclined upper chord bars **Bar 12** and, by symmetry, **Bar 9**.
- **Maximum compressive force and stress:** the axial force is
$$N_{\max} = -51.39\ \text{kN}, \tag{22}$$
and, with $S = 5.0265 \times 10^{-3}$ m^2 ($D = 8$ cm), the corresponding compressive stress is
$$\sigma = \frac{|N_{\max}|}{S} = \frac{51.39 \times 10^3}{5.0265 \times 10^{-3}} \approx 10.22\ \text{MPa}. \tag{23}$$

Note

The vertical end posts (Bars 14 and 7) carry the full support reactions ($N = -50.00$ kN each), but they are short ($L = 2$ m) and slightly less loaded than Bars 12 and 9. The inclined upper chord Bars 12 and 9 therefore govern the compressive resistance of the structure.

3.4.2. Structural Mechanics Interpretation

The compression in the upper chord is governed by the global bending of the truss and by the variation of its depth:

1. **Global bending couple.** Under the downward joint loads, the simply supported truss behaves like a beam with positive bending moment $M_f(x) > 0$. Equilibrium requires tension in the lower chord and compression in the upper chord.
2. **Lever arm.** For a section cut, the compressive force in the upper chord is approximately
$$|N(x)| \approx \frac{M_f(x)}{z(x)}, \tag{24}$$
where $z(x)$ is the effective lever arm (internal depth) of the truss:
   - At the apex ($x = 9$ m), the bending moment is maximal ($M_f = 225$ kN m), but the truss depth is also maximal ($z = 5.0$ m), which gives a large lever arm.
   - In the panel between $x = 3$ m and $x = 6$ m, the moment about the lower node at $x = 6$ m is $M_f = 50\times6 - 10\times6 - 15\times3 = 195$ kN m, while the depth is only 3 to 4 m. The perpendicular lever arm of Bar 12 with respect to that node is $z = 3.795$ m, hence
$$|N_{12}| = \frac{195}{3.795} \approx 51.4\ \text{kN}. \tag{25}$$
   This less favorable M_f/z ratio explains why Bar 12 (and Bar 9) is the most compressed member.
3. **Inclination of the upper chord.** The upper chord is inclined at $\theta = \arctan(1/3) \approx 18.4°$. Its axial force therefore balances both the horizontal component of the bending couple and part of the vertical shear.

3.5. Question 2.2: Verification of No-Buckling

To check the structure against buckling, the Euler critical stress σ_{cr} is compared with the actual compressive stress σ in the critical member.

Since the bars are pinned at both ends, the buckling length equals the geometric length of the member:

$$L_f = L_0. \tag{26}$$

The slenderness ratio is

$$\lambda = \frac{L_f}{i} = \frac{L_0}{0.02}. \tag{27}$$

The Euler critical stress is

$$\sigma_{cr} = \frac{\pi^2 E}{\lambda^2}. \tag{28}$$

The actual compressive stress is

$$\sigma = \frac{|N_{\max}|}{S}. \tag{29}$$

Parameter	Formula	Value
Bar length	$L_0 = \sqrt{3^2 + 1^2}$	3.162 m
Buckling length	$L_f = 1.0 \times L_0$	3.162 m
Section inertia	$I = \frac{\pi D^4}{64}$	2.011×10^{-6} m^4
Cross-section area	$S = \frac{\pi D^2}{4}$	5.027×10^{-3} m^2
Radius of gyration	$i = \sqrt{I/S} = D/4$	0.020 m
Slenderness ratio	$\lambda = L_f/i$	158.11
Critical stress	$\sigma_{cr} = \frac{\pi^2 E}{\lambda^2}$	82.90 MPa
Applied stress	$\sigma = \lvert N_{\max}\rvert / S$	10.22 MPa

Table 2. Euler buckling verification for Bars 12 and 9.

Comparison between σ_{cr} and σ:

$$\sigma = 10.22 \text{ MPa} < \sigma_{cr} = 82.90 \text{ MPa} \qquad \left(\frac{\sigma_{cr}}{\sigma} \approx 8.11\right). \tag{30}$$

Additional check. The end posts (Bars 14 and 7) are the other highly compressed members: $N = -50$ kN, so $\sigma \approx 9.95$ MPa. With $L_0 = 2$ m, $\lambda = 100$ and $\sigma_{cr} = \pi^2 E/\lambda^2 \approx 207$ MPa, so they are also safe against buckling.

Conclusion. The Euler formula is applicable since λ is larger than the limit slenderness of mild steel (≈ 92). The compressive stress in the most loaded bar is more than 8 times smaller than the critical Euler stress, so **no buckling** of any member occurs under the applied loading.

4. EXERCISE 3: BUCKLING CALCULATION OF A THIN PLATE

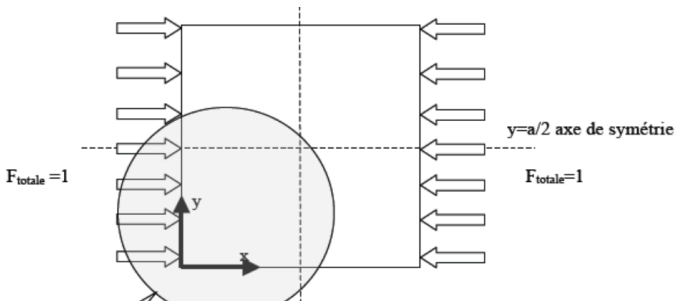


Figure 6. Study model

4.1. Structural Definition and Problem Data

We consider a thin square plate subjected to uniform in-plane uniaxial compression along two opposite parallel edges:

- Plate dimensions: $a = b = 1.0$ m
- Uniform thickness: $h = 5$ mm $= 0.005$ m
- Total compressive load: $F_{\text{total}} = 1.0$ N (unit reference load)
- Material: Aluminum alloy with $E = 70$ GPa $= 70 \times 10^9$ Pa (or 70 MPa as given in test document text), $\nu = 0.33$, and mass density $\rho = 2700$ kg/m^3.
- Global boundary conditions: Simply supported along all four perimeter edges ($T_z = 0$).

4.2. Question 3.1: Symmetry Model and Kinematic Restraints

Owing to the double geometric and loading symmetry about the orthogonal mid-planes $x = a/2$ and $y = b/2$, only one-quarter (1/4) of the plate is modeled ($L_x = 0.5$ m, $L_y = 0.5$ m), substantially reducing computational degrees of freedom.

For a quarter-plate model defined in the (x, y) plane, mirror symmetry conditions require enforcing zero normal displacements and zero rotations about axes lying within the cutting planes:

- **Symmetry plane $x = a/2$ (normal to $\vec{x}$):** Kinematic continuity across the cut requires zero out-of-plane material displacement, enforcing $T_x = 0$. Since the out-of-plane deflection slope along $\vec{x}$ must reach an extremum at mid-span ($\partial w/\partial x = 0$), the bending rotation about the cut line vanishes ($R_y = 0$). Additionally, in-plane drilling rotation is locked ($R_z = 0$) to preserve membrane symmetry and prevent anti-symmetric in-plane shear distortion.
- **Symmetry plane $y = b/2$ (normal to $\vec{y}$):** Similarly, normal translation is constrained ($T_y = 0$), the transverse deflection slope cancels out at the centerline ($\partial w/\partial y = 0 \Rightarrow R_x = 0$), and the in-plane drilling rotation is restrained ($R_z = 0$).

This quarter-model reduction is strictly valid because the fundamental buckling mode of a simply supported rectangular plate under uniaxial compression exhibits a symmetric half-sine wave shape in both directions:

$$w(x, y) = w_0 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi y}{b}\right)$$

Because this mode shape is even with respect to both central axes $x = a/2$ and $y = b/2$, the applied boundary conditions naturally satisfy the mode kinematics without eliminating or artificially stiffening the lowest eigensolution.

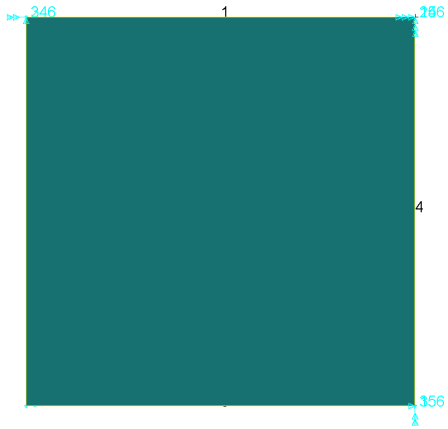


Figure 7. Kinematics conditions

4.3. Mesh Discretization and Consistent Nodal Forces

The quarter-plate is meshed using a regular 6×6 array of 2D QUAD4 shell elements (comprising 36 elements and 49 nodes).

The total compressive load applied to the quarter-plate is:

$$F_{\text{quarter}} = \frac{F_{\text{total}}}{2} = 0.5 \text{ N} \tag{31}$$

Along the loaded edge subdivided into 6 equal element segments, the consistent work-equivalent nodal force distribution follows from the linear shape functions:

- 5 interior nodes each receive a tributary length $\Delta L = \frac{0.5}{6}$ m:

$$F_{\text{int}} = \frac{F_{\text{quarter}}}{6} \approx \boxed{0.0833 \text{ N}} \tag{32}$$

- 2 boundary/corner nodes each receive half a tributary length $\frac{\Delta L}{2} = \frac{0.5}{12}$ m:

$$F_{\text{corner}} = \frac{F_{\text{quarter}}}{12} \approx \boxed{0.0417 \text{ N}} \tag{33}$$

- Consistency verification:

$$5 \times 0.08333 + 2 \times 0.04167 = 0.41665 + 0.08334 = 0.5000 \text{ N} \tag{34}$$

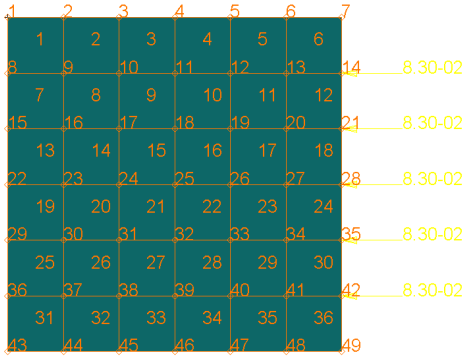
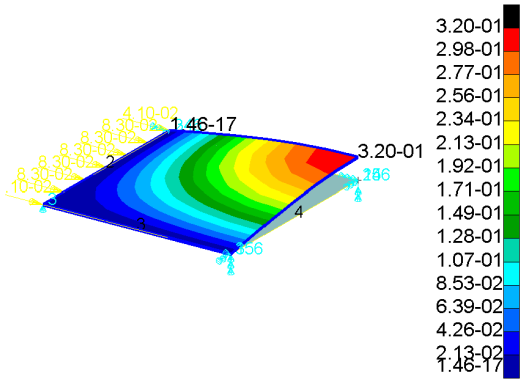


Figure 8. QUAD4 Mesh

4.4. Question 3.2: Finite Element Calculation (SOL 105)

An eigenvalue buckling calculation is conducted using MSC Nastran SOL 105. The primary eigenvalue λ_1 corresponds directly to the critical buckling multiplier:

$$F_{cr,\text{FE}} = \lambda_1 \cdot F_{\text{total}} \tag{35}$$



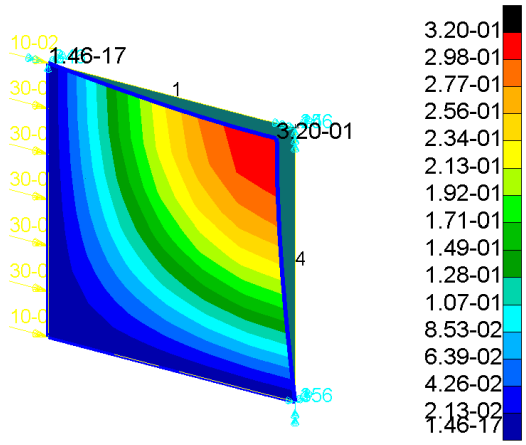


Figure 9. Fundamental buckling mode shape of the simply supported plate

4.5. Question 3.3: Comparison and Error Analysis

The theoretical critical buckling load for a simply supported square plate under uniaxial compression is given by thin plate theory (Kirchhoff-Love):

$$F_{cr,\text{analytical}} = \frac{D\pi^2}{a^2}\left(i + \frac{1}{i}\right)^2 \tag{36}$$

where $a = 1.0$ m is the plate width, and D denotes the plate flexural rigidity:

$$D = \frac{Eh^3}{12(1-\nu^2)} = \frac{70 \times 10^9 \times (5 \times 10^{-3})^3}{12(1-0.33^2)} \approx 818.28 \text{ N} \cdot \text{m} \tag{37}$$

For the fundamental buckling mode ($i = 1$):

$$F_{cr,\text{analytical}} = \frac{4\pi^2 D}{a^2} \approx \boxed{32304.3 \text{ N}} \tag{38}$$

From the MSC Nastran SOL 105 eigensolution, the extracted eigenvalue factor for Mode 1 is $\lambda_1 = 32392.6$. Since a unit reference load ($F_{\text{total}} = 1.0$ N) is distributed via work-equivalent nodal forces, the numerical critical load is:

$$F_{cr,\text{FE}} = \lambda_1 \cdot F_{\text{total}} = \boxed{32392.6 \text{ N}} \tag{39}$$

The relative discrepancy between the theoretical prediction and the finite element eigensolution is computed as:

$$\text{Error } (\%) = \frac{|32304.3 - 32392.6|}{32304.3} \times 100\% \approx \boxed{0.27\%} \tag{40}$$

The finite element model exhibits exceptional agreement with the classical thin plate formulation (0.27% error). The slight overestimation of stiffness is typical of conforming displacement-based shell formulations with discrete mesh discretization.

5. CONCLUSION AND ENGINEERING INSIGHTS

This study highlights the critical role of linear eigenvalue buckling analysis (MSC Nastran SOL 105) in structural sizing: for slender components, catastrophic failure due to elastic instability occurs at stress levels far below the material yield strength, rendering stress-based sizing criteria intrinsically unconservative.

The numerical results demonstrate the extreme sensitivity of critical buckling loads to kinematic boundary conditions: modifying the end constraints of a column, shifting the effective length factor K from 2.0 to 0.7, enhances its load-carrying capacity by a factor of eight. For two-dimensional continuous structures, the consistent enforcement of mirror symmetry constraints on a quarter-plate model substantially reduces computational overhead while capturing the classical Kirchhoff-Love analytical solution with high fidelity (0.27% error).

However, linear eigensolution methods assume geometrically perfect structures and linear pre-buckling kinematics. Consequently, they do not account for the destabilizing effect of initial geometric imperfections, nor do they evaluate the post-buckling reserve strength characteristic of thin panels and stiffened shells. Capturing these physical phenomena requires incremental geometrically nonlinear static formulations (SOL 106).

■ ANNEX: COMPUTATIONAL SCRIPTS

```

import matplotlib.pyplot as plt
import numpy as np

# Typography and plot styling configuration
plt.rcParams.update({
    "font.family": "sans-serif",
    "font.size": 9,
    "axes.labelsize": 10,
    "axes.titlesize": 11,
    "xtick.labelsize": 8.5,
    "ytick.labelsize": 8.5,
    "legend.fontsize": 8.5,
    "figure.titlesize": 13,
    "lines.linewidth": 1.8,
    "grid.alpha": 0.35,
    "grid.linestyle": "--"
})

fig, axs = plt.subplots(2, 2, figsize=(12, 9), dpi=300)
fig.suptitle(
    "IPSA CAE3  TP3: Structural Stability & Buckling Benchmarks (
      ↪ Analytical vs. SOL 105)",
    fontweight="bold", y=0.98
)

# ------------------------------------------------------------------------
      ↪
# PANEL 1: EXERCISE 1 - EULER COLUMN (Effect of K vs. Plastic Yielding)
# ------------------------------------------------------------------------
      ↪
ax1 = axs[0, 0]
L1 = 4.0
D1 = 0.20
e1 = 0.02
d1 = D1 - 2 * e1
E1 = 200e9
sigma_y1 = 600e6

S1 = (np.pi / 4.0) * (D1**2 - d1**2)
I1 = (np.pi / 64.0) * (D1**4 - d1**4)
F_plas1 = (sigma_y1 * S1) / 1e6  # MN

K_cont = np.linspace(0.4, 2.5, 300)
P_cr_cont = (np.pi**2 * E1 * I1) / ((K_cont * L1)**2) / 1e6  # MN

ax1.plot(
    K_cont, P_cr_cont,
    color="#1f77b4",
    label=r"Euler $P_{\mathrm{cr}}(K) \propto 1/K^2$"
)
ax1.axhline(
    F_plas1,
    color="#d62728",
    linestyle="--",
    label=f"$F_{{\\mathrm{{plastic}}}}$ = {F_plas1:.2f} MN (Limit)"
)

cases_K = [2.0, 1.0, 0.7]
cases_labels = [
    "Case 1 (Fixed/Free)",
    "Case 2 (Pinned/Pinned)",
    "Case 3 (Fixed/Pinned)"
]
cases_Pcr = [
    (np.pi**2 * E1 * I1) / ((k * L1)**2) / 1e6
    for k in cases_K
]
colors_pts = ["#2ca02c", "#ff7f0e", "#9467bd"]

for k_val, p_val, lbl, c in zip(
    cases_K, cases_Pcr, cases_labels, colors_pts
):
    ax1.scatter(k_val, p_val, color=c, s=55, zorder=5)
    ax1.annotate(
        f"{lbl}\n{p_val:.2f} MN",
        xy=(k_val, p_val),
        xytext=(10, 8),
        textcoords="offset points",
        fontsize=7.5,
        bbox=dict(
            boxstyle="round,pad=0.2",
            fc="white", ec=c, lw=0.8
        )
    )

ax1.set_title(r"Ex. 1: Column Buckling vs. Boundary Restraint $K$")
```

```python
ax1.set_xlabel("Effective Length Factor $K$")
ax1.set_ylabel("Critical Load [MN]")
ax1.set_yscale("log")
ax1.grid(True, which="both")
ax1.legend(loc="upper right")

# ---------------------------------------------------------------------------
# PANEL 2: EXERCISE 2 - TRUSS (Euler Curve and Plasticity Threshold)
# ---------------------------------------------------------------------------
ax2 = axs[0, 1]
E2 = 210e9
sigma_y2 = 250e6
D_bar = 0.08
i_bar = D_bar / 4.0  # Radius of gyration = 0.02 m

lambda_crit = np.pi * np.sqrt(E2 / sigma_y2)
lambda_range = np.linspace(15, 200, 300)
sigma_euler = (np.pi**2 * E2) / (lambda_range**2) / 1e6  # MPa
sigma_capacity = np.minimum(sigma_euler, sigma_y2 / 1e6)

ax2.plot(
    lambda_range, sigma_euler,
    ":",
    color="gray",
    label=r"Theoretical Euler curve ($\infty$ as $\lambda \to 0$)"
)
ax2.plot(
    lambda_range, sigma_capacity,
    color="#e6550d",
    lw=2.2,
    label=r"Allowable stress curve $(\sigma_{\mathrm{allow}})$"
)
ax2.axvline(
    lambda_crit,
    color="black",
    linestyle="--",
    lw=1,
    label=rf"Critical slenderness $\lambda_{{\mathrm{{crit}}}} = {lambda_crit:.1f}$"
)

# Critical truss member location
L_bar_example = 2.5
lambda_bar = L_bar_example / i_bar
sigma_crit_bar = (np.pi**2 * E2) / (lambda_bar**2) / 1e6

ax2.scatter(
    lambda_bar, sigma_crit_bar,
    color="#756bb1", s=65, zorder=5
)
ax2.annotate(
    f"Truss member ($\\lambda={lambda_bar:.0f}$)\n"
    f"$\\sigma_{{cr}}={sigma_crit_bar:.1f}$ MPa",
    xy=(lambda_bar, sigma_crit_bar),
    xytext=(12, 10),
    textcoords="offset points",
    fontsize=7.5,
    bbox=dict(
        boxstyle="round,pad=0.2",
        fc="white", ec="#756bb1", lw=0.8
    )
)

ax2.set_title(
    r"Ex. 2: Truss Stability Domain "
    r"($\sigma_{\mathrm{cr}}$ vs. Slenderness $\lambda$)"
)
ax2.set_xlabel(r"Slenderness Ratio $\lambda = L_e / i$")
ax2.set_ylabel("Critical Compressive Stress [MPa]")
ax2.set_ylim(0, (sigma_y2 / 1e6) * 1.25)
ax2.grid(True)
ax2.legend(loc="upper right")

# ---------------------------------------------------------------------------
# PANEL 3: EXERCISE 3 - THIN PLATE (h^3 Sensitivity and SOL 105 Result)
# ---------------------------------------------------------------------------
ax3 = axs[1, 0]
a3 = 1.0
nu3 = 0.33
E3 = 70e9
h_nominal = 0.005
FE_result = 32392.6  # N

h_arr = np.linspace(0.002, 0.008, 200)
D_arr = (E3 * h_arr**3) / (12.0 * (1.0 - nu3**2))
F_cr_arr = (4.0 * np.pi**2 * D_arr) / (a3**2) / 1e3  # kN

D_nom = (E3 * h_nominal**3) / (12.0 * (1.0 - nu3**2))
F_cr_nom = (4.0 * np.pi**2 * D_nom) / (a3**2) / 1e3

ax3.plot(
    h_arr * 1e3, F_cr_arr,
    color="#31a354",
    label=r"Kirchhoff-Love $F_{\mathrm{cr}} \propto h^3$"
)
ax3.scatter(
    h_nominal * 1e3, F_cr_nom,
    color="black", s=40, zorder=4,
    label=f"Analytical (5 mm) = {F_cr_nom:.2f} kN"
)
ax3.scatter(
    h_nominal * 1e3, FE_result / 1e3,
    color="#de2d26", marker="x", s=80, lw=2, zorder=5,
    label=f"Nastran SOL 105 = {FE_result/1e3:.2f} kN"
)

ax3.annotate(
    "FE/Theory discrepancy: +0.27%\n"
    "$\\Delta = +88.3$ N",
    xy=(h_nominal * 1e3, FE_result / 1e3),
    xytext=(-110, 15),
    textcoords="offset points",
    arrowprops=dict(
        arrowstyle="->", color="#de2d26", lw=1
    ),
    fontsize=8,
    bbox=dict(
        boxstyle="round,pad=0.25",
        fc="#fee0d2", ec="#de2d26", lw=0.8
    )
)

ax3.set_title(
    r"Ex. 3: Thin Plate Buckling Sensitivity vs. Thickness $h$"
)
ax3.set_xlabel("Plate Thickness $h$ [mm]")
ax3.set_ylabel(r"Critical Load $F_{\mathrm{cr}}$ [kN]")
ax3.grid(True)
ax3.legend(loc="upper left")

# ---------------------------------------------------------------------------
# PANEL 4: SUMMARY OF SOL 105 VS. ANALYTICAL DISCREPANCIES
# ---------------------------------------------------------------------------
ax4 = axs[1, 1]

categories = [
    "Ex. 1 Case 1\n(Fixed/Free)",
    "Ex. 1 Case 2\n(Pinned/Pinned)",
    "Ex. 1 Case 3\n(Fixed/Pinned)",
    "Ex. 3 Plate\n(Quarter Model)"
]
errors = [0.32, 1.28, 2.59, 0.27]
bar_colors = ["#2ca02c", "#ff7f0e", "#9467bd", "#31a354"]

bars = ax4.bar(
    categories, errors,
    color=bar_colors,
    width=0.55,
    edgecolor="black",
    linewidth=0.8
)

for bar, err in zip(bars, errors):
    yval = bar.get_height()
    ax4.text(
        bar.get_x() + bar.get_width() / 2.0,
        yval + 0.08,
        f"{err:.2f}%",
        ha="center",
        va="bottom",
        fontweight="bold",
        fontsize=9
    )

ax4.axhline(0.0, color="black", lw=1)
ax4.set_ylim(0, 3.2)
ax4.set_title("Summary: Relative Discrepancy |FE - Analytical|")
ax4.set_ylabel("Relative Error [%]")
ax4.grid(True, axis="y")

ax4.text(
    0.5, 0.58,
    "SOL 105 Validation:\n"
    " Maximum error < 2.6% across the entire practical session\n"
    " Numerical stiffening is typical of displacement-based formulations\n "
```

```
    " Convergence confirmed using a quarter-plate model (symmetry)",
    transform=ax4.transAxes,
    ha="center",
    va="center",
    fontsize=8,
    bbox=dict(
        boxstyle="round,pad=0.4",
        fc="#f7f7f7",
        ec="#cccccc"
    )
)

plt.tight_layout()
plt.subplots_adjust(top=0.92)

plt.savefig(
    "tp3_stability_benchmarks.png",
    dpi=300,
    bbox_inches="tight"
)
print("--> Plot exported: tp3_stability_benchmarks.png (300 DPI)")
plt.show()
```

Code 1. Python verification script for TP3 analytical stability benchmarks

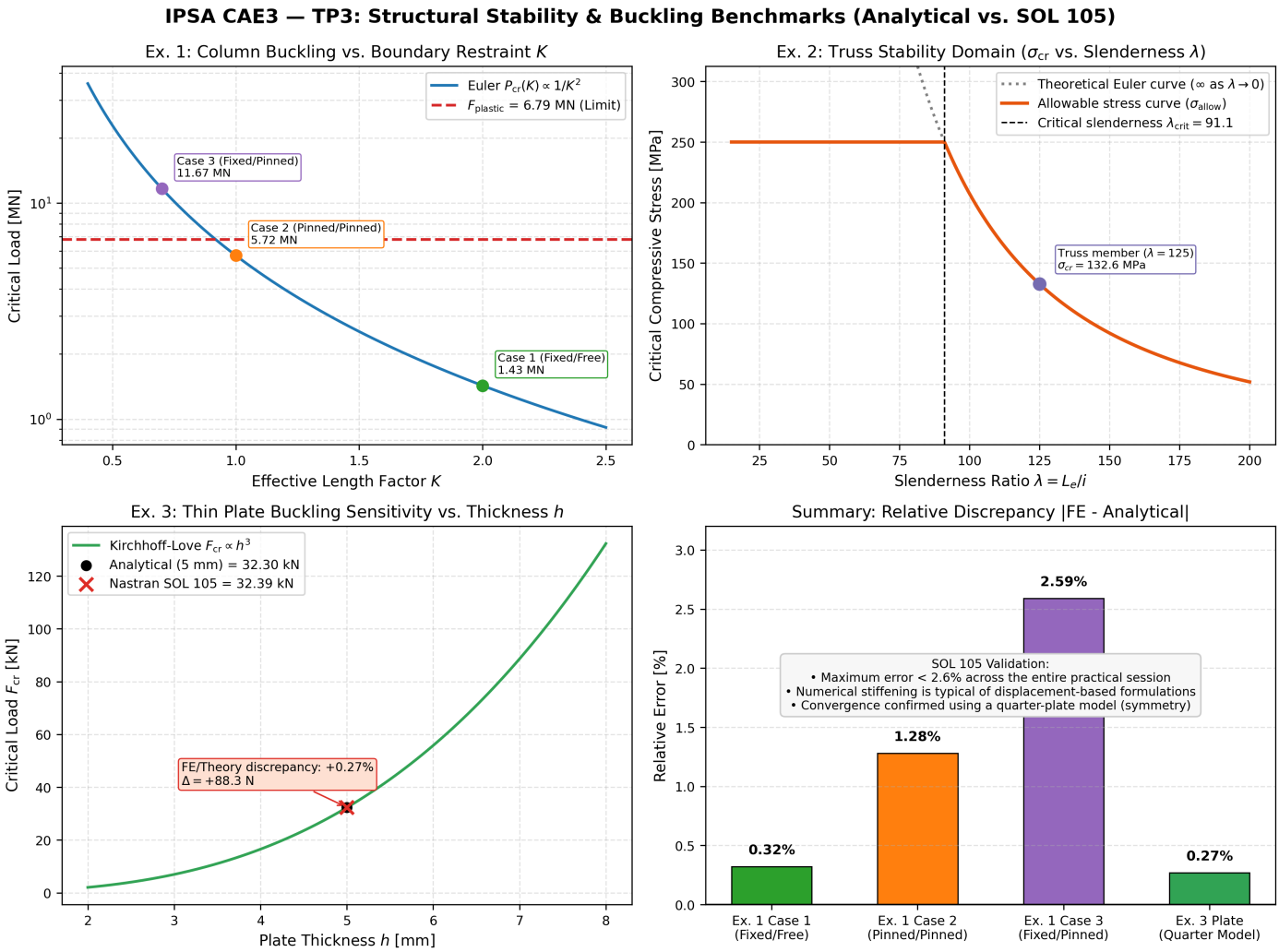


Figure 10. Data summary